```
# dimensionality reduction - PCA.
from sklearn import datasets  # to retrieve the iris Dataset
import pandas as pd  # to load the dataframe
from sklearn.preprocessing import StandardScaler  # to standardize the features
from sklearn.decomposition import PCA  # to apply PCA
import seaborn as sns  # to plot the heat maps
```

```
#Load the Dataset
iris = datasets.load_iris()
#convert the dataset into a pandas data frame
df = pd.DataFrame(iris['data'], columns = iris['feature_names'])
#display the head (first 5 rows) of the dataset
df.head()
```

| | sepal length (cm) | sepal width (cm) | petal length (cm) | petal width (cm) |
|---|---|---|---|---|
| **0** | 5.1 | 3.5 | 1.4 | 0.2 |
| **1** | 4.9 | 3.0 | 1.4 | 0.2 |
| **2** | 4.7 | 3.2 | 1.3 | 0.2 |
| **3** | 4.6 | 3.1 | 1.5 | 0.2 |
| **4** | 5.0 | 3.6 | 1.4 | 0.2 |

Next steps: Generate code with df New interactive sheet

```
#Standardize the features
#Create an object of StandardScaler which is present in sklearn.preprocessing
scalar = StandardScaler()
scaled_data = pd.DataFrame(scalar.fit_transform(df)) #scaling the data
scaled_data
```

| | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| **0** | -0.900681 | 1.019004 | -1.340227 | -1.315444 |
| **1** | -1.143017 | -0.131979 | -1.340227 | -1.315444 |
| **2** | -1.385353 | 0.328414 | -1.397064 | -1.315444 |
| **3** | -1.506521 | 0.098217 | -1.283389 | -1.315444 |
| **4** | -1.021849 | 1.249201 | -1.340227 | -1.315444 |
| **...** | ... | ... | ... | ... |
| **145** | 1.038005 | -0.131979 | 0.819596 | 1.448832 |
| **146** | 0.553333 | -1.282963 | 0.705921 | 0.922303 |
| **147** | 0.795669 | -0.131979 | 0.819596 | 1.053935 |
| **148** | 0.432165 | 0.788808 | 0.933271 | 1.448832 |
| **149** | 0.068662 | -0.131979 | 0.762758 | 0.790671 |

150 rows × 4 columns

Next steps: Generate code with scaled_data New interactive sheet

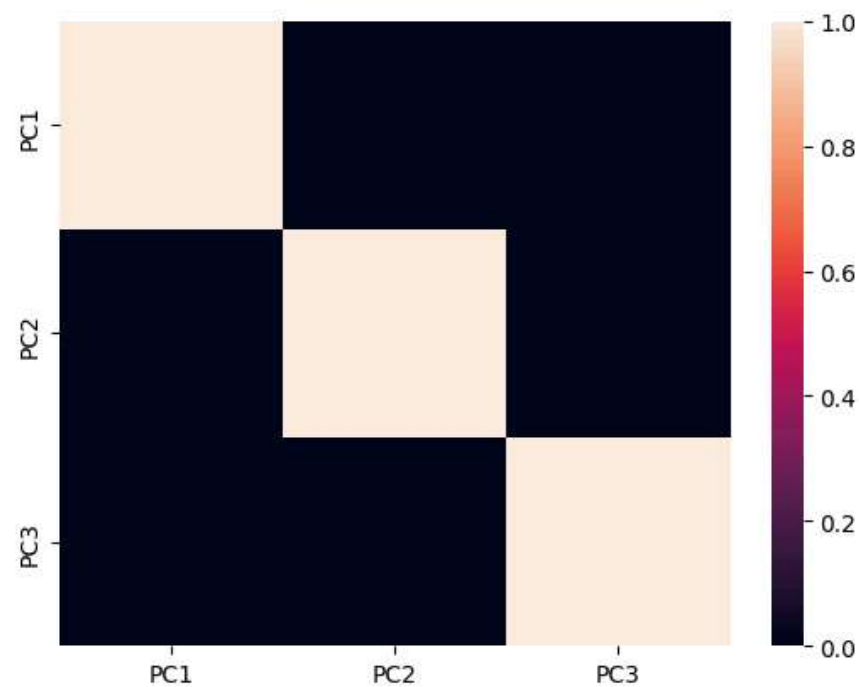
```
#Applying PCA
#Taking no. of Principal Components as 3
pca = PCA(n_components = 3)
pca.fit(scaled_data)
data_pca = pca.transform(scaled_data)
data_pca = pd.DataFrame(data_pca,columns=['PC1','PC2','PC3'])
data_pca.head()
```

| | PC1 | PC2 | PC3 |
|---|---|---|---|
| 0 | -2.264703 | 0.480027 | 0.127706 |
| 1 | -2.080961 | -0.674134 | 0.234609 |
| 2 | -2.364229 | -0.341908 | -0.044201 |
| 3 | -2.299384 | -0.597395 | -0.091290 |
| 4 | -2.389842 | 0.646835 | -0.015738 |

Next steps: Generate code with data_pca New interactive sheet

```
#Checking Co-relation between features after PCA
sns.heatmap(data_pca.corr())
```

```
<Axes: >
```

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